

SAVITRIBAI PHULE PUNE UNIVERSITY

A PROJECT REPORT ON

PROJECT TITLE

**SUBMITTED TOWARDS THE
PARTIAL FULFILLMENT OF THE REQUIREMENTS OF**

BACHELOR OF ENGINEERING (Computer Engineering)

BY

Student Name	Exam No:
Student Name	Exam No:
Student Name	Exam No:
Student Name	Exam No:

Under The Guidance of

Prof. Guide Name

College Logo Here

DEPARTMENT OF COMPUTER ENGINEERING

College Name

College Address

College Logo Here

College Name
DEPARTMENT OF COMPUTER ENGINEERING

CERTIFICATE

This is to certify that the Project Entitled

PROJECT TITLE

Submitted by

Student Name	Exam No:
Student Name	Exam No:
Student Name	Exam No:
Student Name	Exam No:

is a bonafide work carried out by Students under the supervision of Prof. Guide Name and it is submitted towards the partial fulfillment of the requirement of Bachelor of Engineering (Computer Engineering).

Prof. Guide Name
Internal Guide
Dept. of Computer Engg.

Prof. HOD Name
H.O.D
Dept. of Computer Engg.

Dr. V. V. Dixit

Principal

RMD Sinhgad School of Engineering

Signature of Internal Examiner

Signature of External Examiner

PROJECT APPROVAL SHEET

A Project Title

(Project Title)

Is successfully completed by

Student Name

Exam No:

Student Name

Exam No:

Student Name

Exam No:

Student Name

Exam No:

at

DEPARTMENT OF COMPUTER ENGINEERING

(COLLEGE NAME)

SAVITRIBAI PHULE PUNE UNIVERSITY,PUNE

ACADEMIC YEAR 2019-2020

Prof. Guide Name

Prof. HOD Name

Internal Guide

H.O.D

Dept. of Computer Engg.

Dept. of Computer Engg.

Abstract

Please Write here One Page Abstract. It should mainly include introduction, motivation, outcome and innovation if any.

Acknowledgments

Please Write here Acknowledgment.Example given as

*It gives us great pleasure in presenting the preliminary project report on '**BE PROJECT TITLE**'.*

*I would like to take this opportunity to thank my internal guide **Prof. Guide Name** for giving me all the help and guidance I needed. I am really grateful to them for their kind support. Their valuable suggestions were very helpful.*

*I am also grateful to **Prof. HOD Name**, Head of Computer Engineering Department, CollegeName for his indispensable support, suggestions.*

*In the end our special thanks to **Other Person Name** for providing various resources such as laboratory with all needed software platforms, continuous Internet connection, for Our Project.*

Student Name1

Student Name2

Student Name3

Student Name4

(B.E. Computer Engg.)

INDEX

1	Synopsis	1
1.1	Project Title	2
1.2	Project Option	2
1.3	Internal Guide	2
1.4	Sponsorship and External Guide	2
1.5	Technical Keywords (As per ACM Keywords)	2
1.6	Problem Statement	2
1.7	Abstract	3
1.8	Goals and Objectives	3
1.9	Relevant mathematics associated with the Project	3
1.10	Names of Conferences / Journals where papers can be published . .	3
1.11	Review of Conference/Journal Papers supporting Project idea	4
1.12	Plan of Project Execution	4
2	Technical Keywords	5
2.1	Area of Project	6
2.2	Technical Keywords	6
3	Introduction	7
3.1	Project Idea	8
3.2	Motivation of the Project	8
3.3	Literature Survey	8
4	Problem Definition and scope	9
4.1	Problem Statement	10
4.1.1	Goals and objectives	10
4.1.2	Statement of scope	10

4.2	Major Constraints.....	10
4.3	Methodologies of Problem solving and efficiency issues.....	10
	4.4 Outcome	10
4.5	Applications	11
4.6	Hardware Resources Required.....	11
4.7	Software Resources Required.....	11
5	Project Plan	12
5.1	Project Estimates.....	13
5.1.1	Reconciled Estimates	13
5.1.2	Project Resources.....	13
5.2	Risk Management w.r.t. NPHard analysis	13
5.2.1	Risk Identification.....	13
	5.2.2 Risk Analysis	14
5.2.3	Overview of Risk Mitigation, Monitoring, Management	15
5.3	Project Schedule.....	15
	5.3.1 Project task set	15
	5.3.2 Task network	16
	5.3.3 Timeline Chart	16
5.4	Team Organization	16
	5.4.1 Team structure	16
5.4.2	Management reporting and communication	16
6	Software requirement specification	17
6.1	Introduction.....	18
6.1.1	Purpose and Scope of Document	18
6.1.2	Overview of responsibilities of Developer	18
6.2	Usage Scenario.....	18
	6.2.1 User profiles	18
	6.2.2 Use-cases	18

6.2.3	Use Case View	18
6.3	Data Model and Description	19
6.3.1	Data Description	19
6.3.2	Data objects and Relationships	19
6.4	Functional Model and Description	19
6.4.1	Data Flow Diagram	20
6.4.2	Activity Diagram:	20
6.4.3	Non Functional Requirements:	20
	6.4.4 State Diagram:	20
6.4.5	Design Constraints	20
6.4.6	Software Interface Description	20
7	Detailed Design Document using Appendix A and B	22
7.1	Introduction	23
7.2	Architectural Design	23
7.3	Data design (using Appendices A and B)	23
7.3.1	Internal software data structure	23
7.3.2	Global data structure	24
7.3.3	Temporary data structure	24
7.3.4	Database description	24
7.4	Component Design	24
	7.4.1 Class Diagram	24
8	Project Implementation	26
8.1	Introduction	27
8.2	Tools and Technologies Used	27
8.3	Methodologies/Algorithm Details	27
8.3.1	Algorithm 1/Pseudo Code	27
8.3.2	Algorithm 2/Pseudo Code	27
8.4	Verification and Validation for Acceptance	27
9	Software Testing	28
9.1	Type of Testing Used	29

9.2 Test Cases and Test Results.....	29
10 Results	30
10.1 Screen shots	31
10.2 Outputs	31
11 Deployment and Maintenance	32
11.1 Installation and un-installation	33
11.2 User help.....	33
12 Conclusion and future scope	34
Annexure A References	36
Annexure B Project Planner	37
Annexure C Reviewers Comments of Paper Submitted	38
Annexure D Plagiarism Report	39
Annexure E Information of Project Group Members	40

List of Figures

6.1	Use case diagram.....	19
6.2	State transition diagram	21
7.1	Architecture diagram	23
7.2	Class Diagram	25

List of Tables

4.1	Hardware Requirements	11
5.1	Risk Table	14
5.2	Risk Probability definitions [?]	14
5.3	Risk Impact definitions [?]	14
6.1	Use Cases	18
B.1	IDEA Matrix	37

CHAPTER 1

SYNOPSIS

1.1 PROJECT TITLE

Project Title

1.2 PROJECT OPTION

Please mention type either industry sponsored, entrepreneur or internal project

1.3 INTERNAL GUIDE

Prof. Internal Guide Name

1.4 SPONSORSHIP AND EXTERNAL GUIDE

Please write if any sponsorship

1.5 TECHNICAL KEYWORDS (AS PER ACM KEYWORDS)

Please note ACM Keywords can be found : <http://www.acm.org/about/class/ccs98-html>

Example is given as

1. C. Computer Systems Organization
 - (a) C.2 COMPUTER-COMMUNICATION NETWORKS
 - i. C.2.4 Distributed Systems
 - A. Client/server
 - B. Distributed applications
 - C. Distributed databases
 - D. Network operating systems
 - E. Distributed file systems
 - F. Security and reliability issues in distributed applications

1.6 PROBLEM STATEMENT

Define Problem Statement

1.7 ABSTRACT

- Abstract (10 to 15 lines)

1.8 GOALS AND OBJECTIVES

- Objectives

1.9 RELEVANT MATHEMATICS ASSOCIATED WITH THE PROJECT

System Description:

- Input:
- Output:
- Identify data structures, classes, divide and conquer strategies to exploit distributed/parallel/concurrent processing, constraints.
- Functions : Identify Objects, Morphisms, Overloading in functions, Functional relations
- Mathematical formulation if possible
- Success Conditions:
- Failure Conditions:

1.10 NAMES OF CONFERENCES / JOURNALS WHERE PAPERS CAN BE PUBLISHED

- IEEE/ACM Conference/Journal 1
- Conferences/workshops in IITs
- Central Universities or SPPU Conferences
- IEEE/ACM Conference/Journal 2

1.11 REVIEW OF CONFERENCE/JOURNAL PAPERS SUPPORTING PROJECT IDEA

Atleast 10 papers + White papers or web references

Brief literature survey [Description containing important description of at least 10 papers

1.12 PLAN OF PROJECT EXECUTION

Using planner or alike project management tool.

CHAPTER 2

TECHNICAL KEYWORDS

2.1 AREA OF PROJECT

Project Area

2.2 TECHNICAL KEYWORDS

Please note ACM Keywords can be found : <http://www.acm.org/about/class/ccs98.html>

Example is given as

1. C. Computer Systems Organization

(a) C.2 COMPUTER-COMMUNICATION NETWORKS

i. C.2.4 Distributed Systems

- A. Client/server**
- B. Distributed applications**
- C. Distributed databases**
- D. Network operating systems**
- E. Distributed file systems**
- F. Security and reliability issues in distributed applications**

CHAPTER 3

INTRODUCTION

3.1 PROJECT IDEA

- Project Idea

3.2 MOTIVATION OF THE PROJECT

- Motivation of the Project

3.3 LITERATURE SURVEY

- Review of the papers, Description , Mathematical Terms

CHAPTER 4

PROBLEM DEFINITION AND SCOPE

4.1 PROBLEM STATEMENT

Description of Problem

4.1.1 Goals and objectives

Goal and Objectives:

- Overall goals and objectives of software, input and output description with necessary syntax, format etc are described

4.1.2 Statement of scope

- A description of the software with Size of input, bounds on input, input validation, input dependency, i/o state diagram, Major inputs, and outputs are described without regard to implementation detail.
- The scope identifies what the product is and is not, what it will and wont do, what it will and wont contain.

4.2 MAJOR CONSTRAINTS

- Any constraints that will impact the manner in which the software is to be specified, designed, implemented or tested are noted here.

4.3 METHODOLOGIES OF PROBLEM SOLVING AND EFFICIENCY ISSUES

- The single problem can be solved by different solutions. This considers the performance parameters for each approach. Thus considers the efficiency issues.

4.4 OUTCOME

- Outcome of the project

4.5 APPLICATIONS

- Applications of Project

4.6 HARDWARE RESOURCES REQUIRED

Sr. No.	Parameter	Minimum Requirement	Justification
1	CPU Speed	2 GHz	Remark Required
2	RAM	3 GB	Remark Required

Table 4.1: Hardware Requirements

4.7 SOFTWARE RESOURCES REQUIRED

Platform :

1. Operating System:
2. IDE:
3. Programming Language

CHAPTER 5

PROJECT PLAN

5.1 PROJECT ESTIMATES

Use Waterfall model and associated streams derived from assignments 1,2, 3, 4 and 5(Annex A and B) for estimation.

5.1.1 Reconciled Estimates

5.1.1.1 Cost Estimate

5.1.1.2 Time Estimates

5.1.2 Project Resources

Project resources [People, Hardware, Software, Tools and other resources] based on Memory Sharing, IPC, and Concurrency derived using appendices to be referred.

5.2 RISK MANAGEMENT W.R.T. NP HARD ANALYSIS

This section discusses Project risks and the approach to managing them.

5.2.1 Risk Identification

For risks identification, review of scope document, requirements specifications and schedule is done. Answers to questionnaire revealed some risks. Each risk is categorized as per the categories mentioned in [?]. Please refer table 5.1 for all the risks. You can refered following risk identification questionnaire.

1. Have top software and customer managers formally committed to support the project?
2. Are end-users enthusiastically committed to the project and the system/product to be built?
3. Are requirements fully understood by the software engineering team and its customers?
4. Have customers been involved fully in the definition of requirements?
5. Do end-users have realistic expectations?

6. Does the software engineering team have the right mix of skills?
7. Are project requirements stable?
8. Is the number of people on the project team adequate to do the job?
9. Do all customer/user constituencies agree on the importance of the project and on the requirements for the system/product to be built?

5.2.2 Risk Analysis

The risks for the Project can be analyzed within the constraints of time and quality

ID	Risk Description	Probability	Impact		
			Schedule	Quality	Overall
1	Description 1	Low	Low	High	High
2	Description 2	Low	Low	High	High

Table 5.1: Risk Table

Probability	Value	Description
High	Probability of occurrence is	> 75%
Medium	Probability of occurrence is	26 – 75%
Low	Probability of occurrence is	< 25%

Table 5.2: Risk Probability definitions [?]

Impact	Value	Description
Very high	> 10%	Schedule impact or Unacceptable quality
High	5 – 10%	Schedule impact or Some parts of the project have low quality
Medium	< 5%	Schedule impact or Barely noticeable degradation in quality Low Impact on schedule or Quality can be incorporated

Table 5.3: Risk Impact definitions [?]

5.2.3 Overview of Risk Mitigation, Monitoring, Management

Following are the details for each risk.

Risk ID	1
Risk Description	Description 1
Category	Development Environment.
Source	Software requirement Specification document.
Probability	Low
Impact	High
Response	Mitigate
Strategy	Strategy
Risk Status	Occurred

Risk ID	2
Risk Description	Description 2
Category	Requirements
Source	Software Design Specification documentation review.
Probability	Low
Impact	High
Response	Mitigate
Strategy	Better testing will resolve this issue.
Risk Status	Identified

5.3 PROJECT SCHEDULE

5.3.1 Project task set

Major Tasks in the Project stages are:

- Task 1:
- Task 2:
- Task 3:

Risk ID	3
Risk Description	Description 3
Category	Technology
Source	This was identified during early development and testing.
Probability	Low
Impact	Very High
Response	Accept
Strategy	Example Running Service Registry behind proxy balancer
Risk Status	Identified

- Task 4:
- Task 5:

5.3.2 Task network

Project tasks and their dependencies are noted in this diagrammatic form.

5.3.3 Timeline Chart

A project timeline chart is presented. This may include a time line for the entire project. Above points should be covered in Project Planner as Annex C and you can mention here Please refer Annex C for the planner

5.4 TEAM ORGANIZATION

The manner in which staff is organized and the mechanisms for reporting are noted.

5.4.1 Team structure

The team structure for the project is identified. Roles are defined.

5.4.2 Management reporting and communication

Mechanisms for progress reporting and inter/intra team communication are identified as per assessment sheet and lab time table.

CHAPTER 6

SOFTWARE REQUIREMENT

SPECIFICATION

6.1 INTRODUCTION

6.1.1 Purpose and Scope of Document

The purpose of SRS and what it covers is to be stated

6.1.2 Overview of responsibilities of Developer

What all activities carried out by developer?

6.2 USAGE SCENARIO

This section provides various usage scenarios for the system to be developed.

6.2.1 User profiles

The profiles of all user categories are described here.(Actors and their Description)

6.2.2 Use-cases

All use-cases for the software are presented. Description of all main Use cases using use case template is to be provided.

Sr No.	Use Case	Description	Actors	Assumptions
1	Use Case 1	Description	Actors	Assumption

Table 6.1: Use Cases

6.2.3 Use Case View

Use Case Diagram. Example is given below

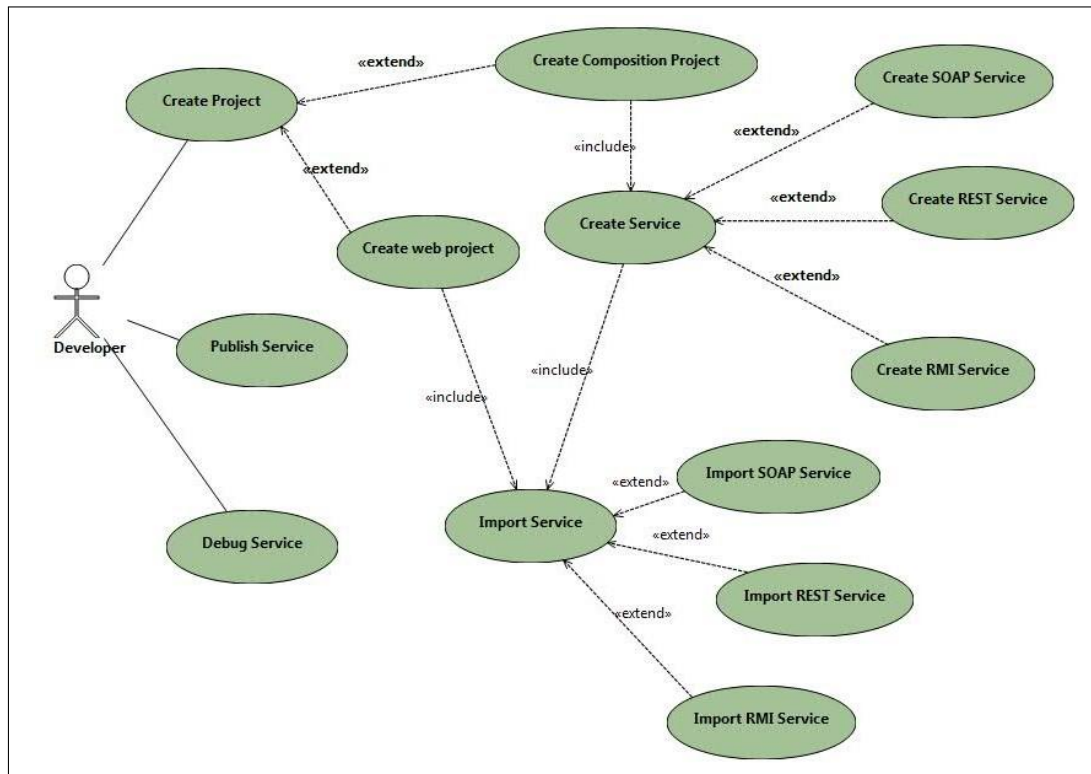


Figure 6.1: Use case diagram

6.3 DATA MODEL AND DESCRIPTION

6.3.1 Data Description

Data objects that will be managed/manipulated by the software are described in this section. The database entities or files or data structures required to be described. For data objects details can be given as below

6.3.2 Data objects and Relationships

Data objects and their major attributes and relationships among data objects are described using an ERD- like form.

6.4 FUNCTIONAL MODEL AND DESCRIPTION

A description of each major software function, along with data flow (structured analysis) or class hierarchy (Analysis Class diagram with class description for object oriented system) is presented.

6.4.1 Data Flow Diagram

6.4.1.1 Level 0 Data Flow Diagram

6.4.1.2 Level 1 Data Flow Diagram

6.4.2 Activity Diagram:

- The Activity diagram represents the steps taken.

6.4.3 Non Functional Requirements:

- Interface Requirements
- Performance Requirements
- Software quality attributes such as availability [related to Reliability], modifiability [includes portability, reusability, scalability] , performance, security, testability and usability[includes self adaptability and user adaptability]

6.4.4 State Diagram:

State Transition Diagram

Fig.6.2 example shows the state transition diagram of Cloud SDK. The states are represented in ovals and state of system gets changed when certain events occur. The transitions from one state to the other are represented by arrows. The Figure shows important states and events that occur while creating new project.

6.4.5 Design Constraints

Any design constraints that will impact the subsystem are noted.

6.4.6 Software Interface Description

The software interface(s) to the outside world is(are) described. The requirements for interfaces to other devices/systems/networks/human are stated.

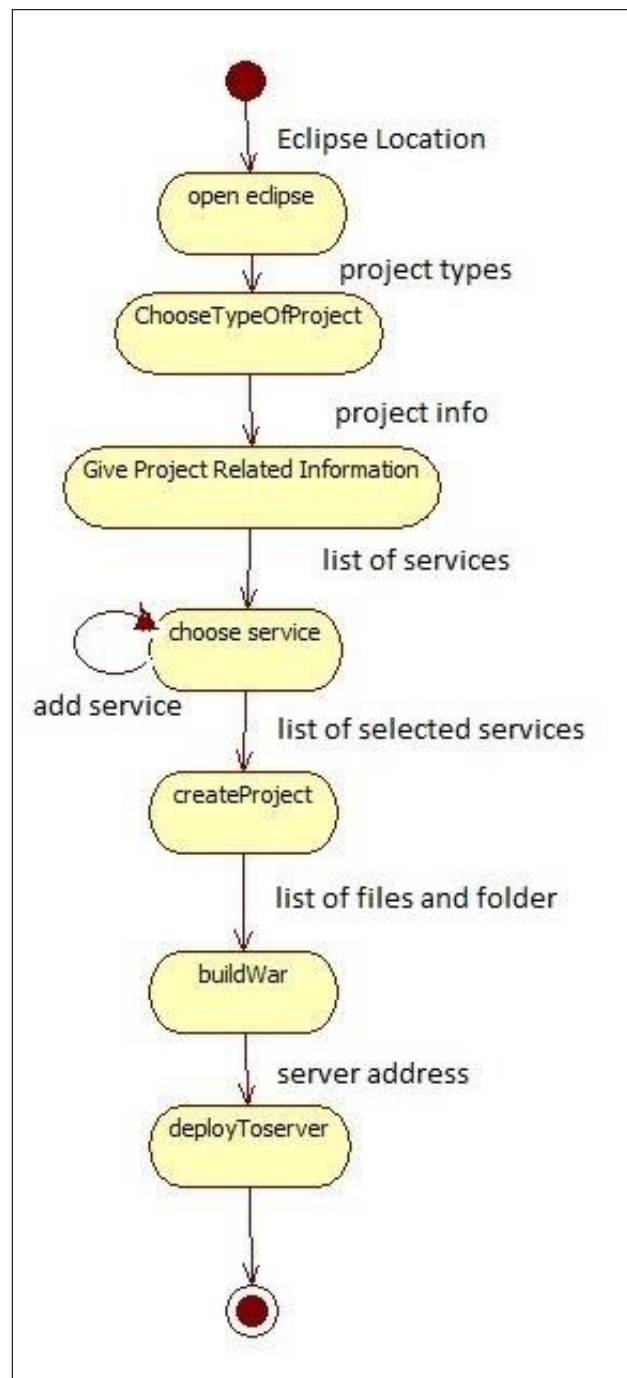


Figure 6.2: State transition diagram

CHAPTER 7

DETAILED DESIGN DOCUMENT USING

APPENDIX A AND B

7.1 INTRODUCTION

This document specifies the design that is used to solve the problem of Product.

7.2 ARCHITECTURAL DESIGN

A description of the program architecture is presented. Subsystem design or Block diagram, Package Diagram, Deployment diagram with description is to be presented.

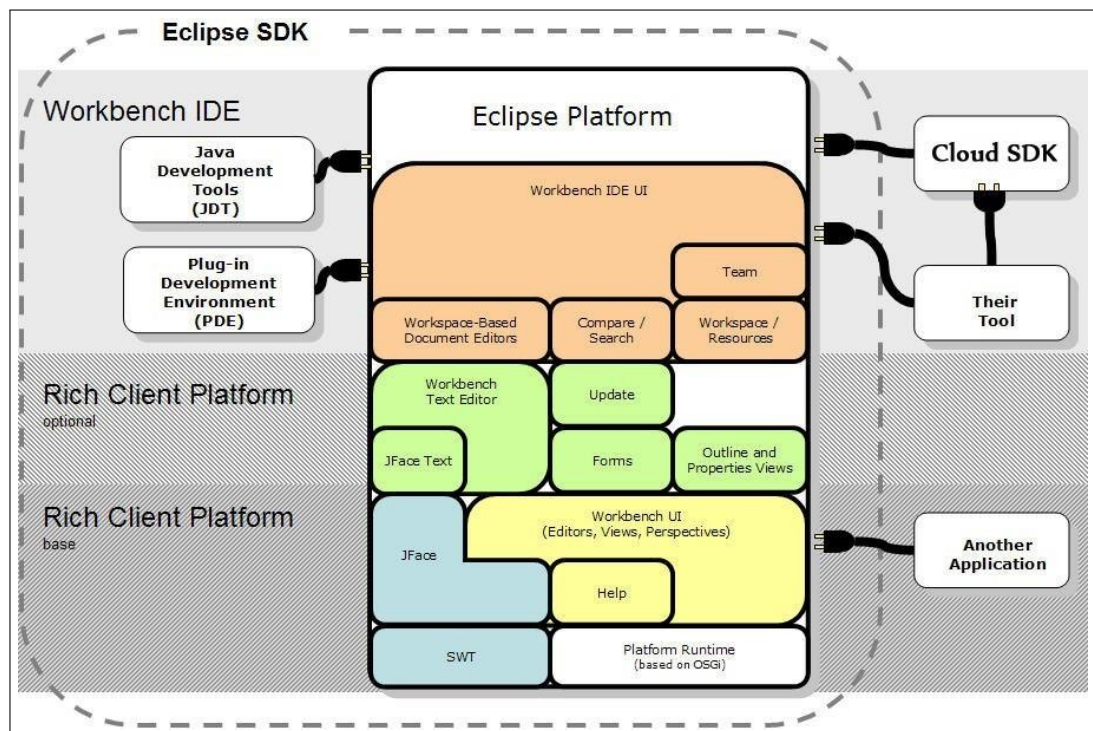


Figure 7.1: Architecture diagram

7.3 DATA DESIGN (USING APPENDICES A AND B)

A description of all data structures including internal, global, and temporary data structures, database design (tables), file formats.

7.3.1 Internal software data structure

Data structures that are passed among components the software are described.

7.3.2 Global data structure

Data structured that are available to major portions of the architecture are described.

7.3.3 Temporary data structure

Files created for interim use are described.

7.3.4 Database description

Database(s) / Files created/used as part of the application is(are) described.

7.4 COMPOENT DESIGN

Class diagrams, Interaction Diagrams, Algorithms. Description of each component description required.

7.4.1 Class Diagram

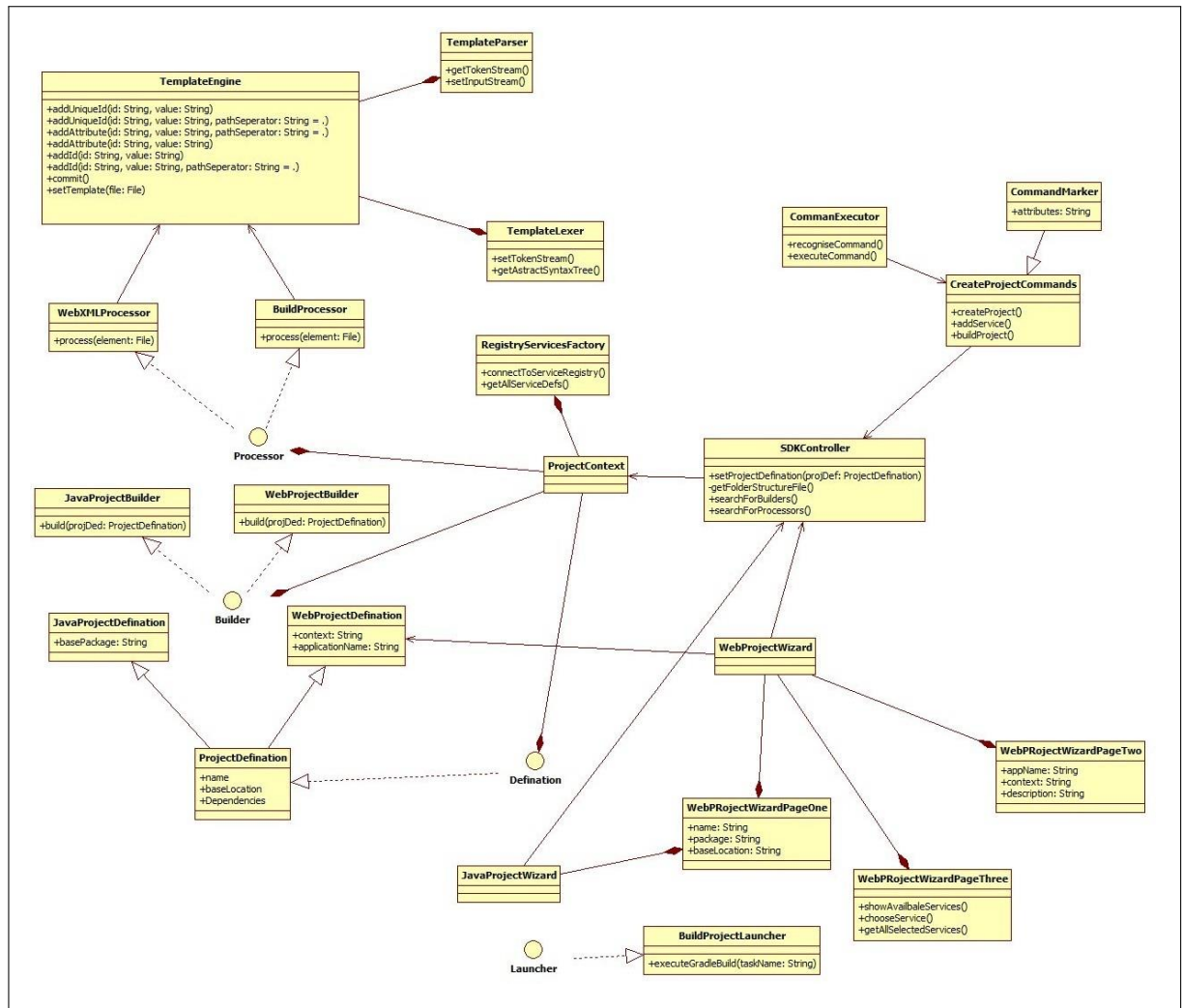


Figure 7.2: Class Diagram

CHAPTER 8

PROJECT IMPLEMENTATION

8.1 INTRODUCTION

8.2 TOOLS AND TECHNOLOGIES USED

8.3 METHODOLOGIES/ALGORITHM DETAILS

8.3.1 Algorithm 1/Pseudo Code

8.3.2 Algorithm 2/Pseudo Code

8.4 VERIFICATION AND VALIDATION FOR ACCEPTANCE

CHAPTER 9

SOFTWARE TESTING

9.1 TYPE OF TESTING USED

Unit,Integration,system etc.

9.2 TEST CASES AND TEST RESULTS

for each type of testing done.

CHAPTER 10

RESULTS

10.1 SCREEN SHOTS

Outputs / Snap shots of the results

10.2 OUTPUTS

Outputs / Snap shots of the results

CHAPTER 11

DEPLOYMENT AND MAINTENANCE

11.1 INSTALLATION AND UN-INSTALLATION

11.2 USER HELP

CHAPTER 12

CONCLUSION AND FUTURE SCOPE

Write summary , conclusion and future scope

ANNEXURE A

REFERENCES

(Strictly in ACM Format)

ANNEXURE B

PROJECT PLANNER

Using planner or alike project management tool.

ANNEXURE C

REVIEWERS COMMENTS OF PAPER

SUBMITTED

(At-least one technical paper must be submitted in Term-I on the project design in the conferences/workshops in IITs, Central Universities or UoP Conferences or equivalent International Conferences Sponsored by IEEE/ACM)

1. Paper Title:
2. Name of the Conference/Journal where paper submitted :
3. Paper accepted/rejected :
4. Review comments by reviewer :
5. Corrective actions if any :

ANNEXURE D

PLAGIARISM REPORT

Plagiarism report

ANNEXURE E

INFORMATION OF PROJECT GROUP

MEMBERS

one page for each student .



photo.jpg

1. Name :
2. Date of Birth :
3. Gender :
4. Permanent Address :
5. E-Mail :
6. Mobile/Contact No. :
7. Placement Details :
8. Paper Published :

